



Building a boundary-spanning service for coopetition



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ABSTRACT

Coopetition is a revolutionary mindset, combining cooperation and competition and allowing companies to work with potential collaborators based on their preferences. It challenges the concept of a supply chain, where companies occupy different positions within a layered structure. Relationships in a supply chain set boundaries for the different collaborators. In contrast, the partners in coopetition can be either collaborators or competitors, and the relationship can evolve over time. To support coopetition, a platform is required for dynamic relationships to develop and for various types of collaboration to occur. This platform enables companies to effectively form networks via a boundary-spanning service, and thus gain competitive advantages. A boundary-spanning service is an intelligent intermediary, providing information, contingencies, and role routines with discretionary powers under constraints. This study demonstrates a process of collaborative purchasing whereby companies negotiate over different offers to increase their utilities using boundary-spanning agents.

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1. Introduction

Collaborative networks allow businesses to extend collaboration agreements beyond the traditional layered supply chain to include various situations and participants. Networks allow for collaborations between members of a supply chain who are directly connected, on the same layer of the chain, or who are further upstream or downstream. Traditionally, forming tighter supply chain relationships results in competitive advantages by enabling the quick responses that are necessary to achieve strategic goals (Daft & Weick, 1984), such as lower inventory levels. Companies on the same layer of a supply chain are normally considered as competitors. The relationships therefore form boundaries to the layered network.

However, when a collaborative network is established, the competitive relationship becomes blurred. A company must maintain good relationships not only with suppliers and buyers, but also with their same-layer competitors. This revolutionary approach is termed coopetition, and is a combination of cooperation and competition (Brandenburger & Nalebuff, 1996). The coopetition relationship can create different types of cooperation, which ultimately benefit a company; for example, a company can work with its same-layer competitors to obtain a better volume discount from a seller.

The establishment of a collaborative network allows the possible modes of cooperation and competition to be extremely dynamic.

Suppliers, buyers, and other collaborators can be easily located, maximizing efficiency. In this study, we examine the use of purchasing data in a cloud-based collaborative network. Boundary-spanning agents allow participants to identify other collaborators and negotiate with them. These boundary-spanning agents—also known as boundary spanners or boundary-spanning individuals—are intelligent intermediaries. Their functions are to (1) process and filter information; (2) provide for contingencies; (3) create the role routine; (4) possess some degree of discretionary power; (5) take on different technological and boundary roles (mediating, an inter-dependent role, or intensive interaction) for different types of organizations; and (6) deal with environmental constraints (Aldrich & Herker, 1977; Ancona & Caldwell, 1992; Tushman & Scanlan, 1981).

In our study we have designed an intelligent, boundary-spanning agent that provides decision-making support and can automatically search for, negotiate, and evaluate solutions for a participant (Barbati, Bruno, & Genovese, 2012). A collaboration support tool is not a type of collaborative software or “groupware” (Ellis, Gibbs, & Rein, 1991) enabling different individuals involved in a common task to achieve their goals. It instead works at the firm level, similar to a negotiation support system (NSS), such as active collaboration and negotiation frameworks (ACNF), where active documents with embedded business rules are used to adapt to different collaborative strategies in a business-to-business (B2B) environment (Du & Chen, 2007). It is also comparable to DSS agents used to reason cooperatively with other agents in a process of interactive constraint satisfaction (Wang, Liao, & Liao, 2002). A collaboration support tool should be more than an NSS, in which the primary concerns are the negotiations with other

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